GENETIC ALGORITHM: A brief introduction

Introduction

- Genetic algorithms are a randomized heuristic search strategy, inspired by Charles Darwin's theory of natural evolution.
- Algorithm reflects the process of natural selection where the fittest individuals are selected for reproduction in order to produce offspring of the next generation.
- It is an optimization technique, which tries to find out such values of input so that we get the best output values or results.
- It's a type of local search that uses methods based on evolution to make small changes to a population of chromosomes in an attempt to identify an optimal solution.

Basic terminologies

- **Population** of individuals
  - Individual is feasible solution to problem
- Each individual is characterized by a **Fitness function**
  - Higher fitness is better solution
- Based on their fitness, parents are selected to reproduce **offspring** for a new **generation**
  - Fitter individuals have more chance to reproduce
  - New generation has same size as old generation; old generation dies
- Offspring has **combination** of properties of two parents
- If well designed, population will **converge** to optimal solution

Representations

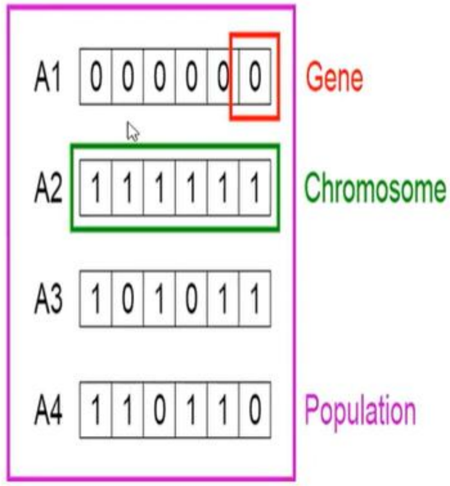
- An individual is characterized by a set of parameters: **Genes.**
  - Each bit is known as a **gene.**
- The genes are joined into a string: **Chromosome.**
  - A string of bits is known as a **chromosome.**
- The chromosome forms the **genotype**
- The genotype contains all information to construct an organism: the **phenotype**
- **Fitness** is measured in the real world ('struggle for life') of the **phenotype**
- Chromosomes can be combined together to form **creatures**

Phases in a Genetic Algorithm

- Initial population
- Fitness function
- Selection
- Crossover
- Mutation

Initial Population

- The process begins with a set of individuals which is called a Population
- An individual is characterized by a set of parameters (variables) known as Genes. Genes are joined into a string to form Chromosome (Solution)
- Usually, binary values are used to encode the genes in a chromosome



Fitness function

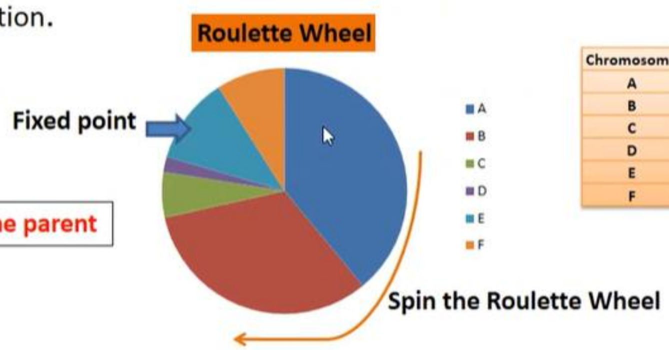
- The **fitness function** determines how fit an individual is (the ability of an individual to compete with other individuals).
- It gives a **fitness score** to each individual.
- The probability that an individual will be selected for reproduction is based on its fitness score.

Fitness Proportionate Selection

- Select fitter individuals for reproduction
- Types of fitness proportionate selection:
  - Roulette Wheel Selection
  - Tournament Selection
  - Rank Selection

Roulette Wheel Selection

Circular wheel is divided into ***n*** pies. where ***n*** is the number of individuals in the population.

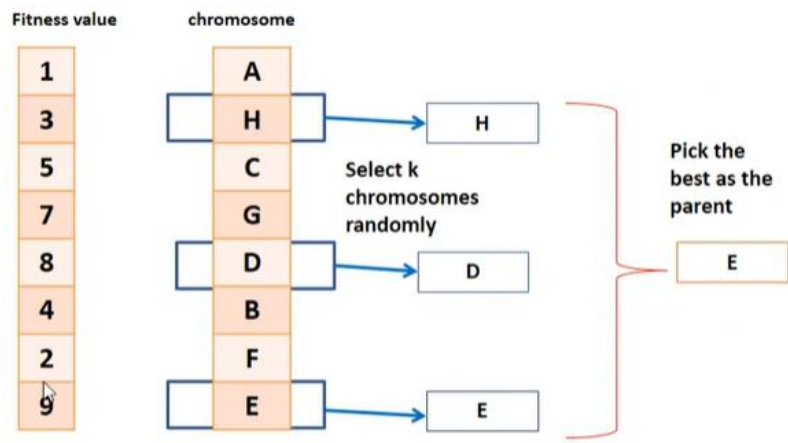


Chromosome	Fitness value
A	7.8
B	6.5
C	1.2
D	0.4
E	2.3
F	1.8

Choose E as the parent

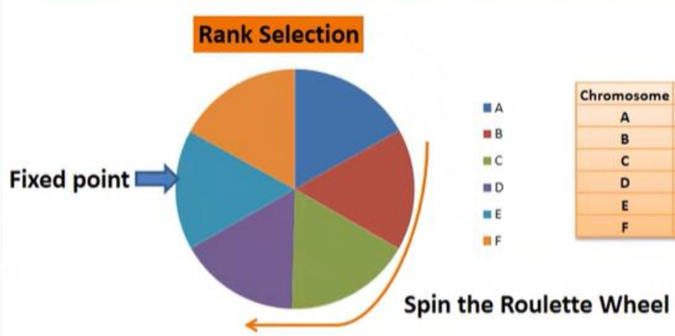
- **A fixed point is chosen on the wheel circumference as shown and the wheel is rotated.**
- **The region of the wheel which comes in front of the fixed point is chosen as the parent.**
- **For the second parent, the same process is repeated.**
- **Fitter individual has a greater pie on the wheel and therefore a greater chance of selectio**

Tournament Selection



- Select K individuals from the population at random and select the best out of these to become a parent.
- The same process is repeated for selecting the next parent.
- Tournament Selection can even work with negative fitness values.

Rank Selection



Chromosome	Fitness value
A	7.1
B	6.88
C	7.2
D	6.9
E	6.8
F	7.15

Chromosome	Fitness value	Rank
A	7.1	3
B	6.88	5
C	7.2	1
D	6.9	4
E	6.8	6
F	7.15	2

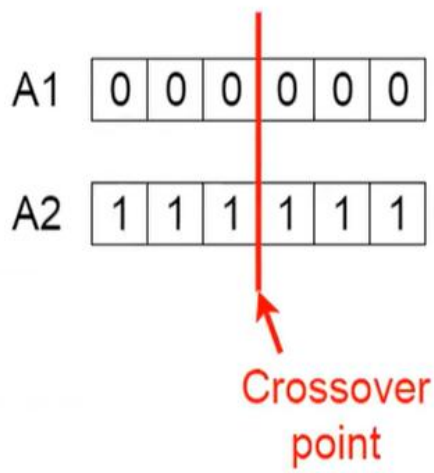
- Individuals in the population have very close fitness values
- Each individual having an almost equal share of the pie
- Each individual no matter how fit relative to each other has an approximately same probability of getting selected as a parent
- Concept of a fitness value is removed while selecting a parent.
- Every individual in the population is ranked according to their fitness
- The selection of the parents depends on the rank of each individual and not the fitness.
- The higher ranked individuals are preferred more than the lower ranked ones.

Selection

- The idea of **selection** phase is to select the fittest individuals and let them pass their genes to the next generation.
- Two pairs of individuals (**parents**) are selected based on their fitness scores.
- Individuals with high fitness have more chance to be selected for reproduction.

Crossover

- For each pair of parents to be mated, a **crossover point** is chosen at random from within the genes.
- For example, consider the crossover point to be 3 as shown.

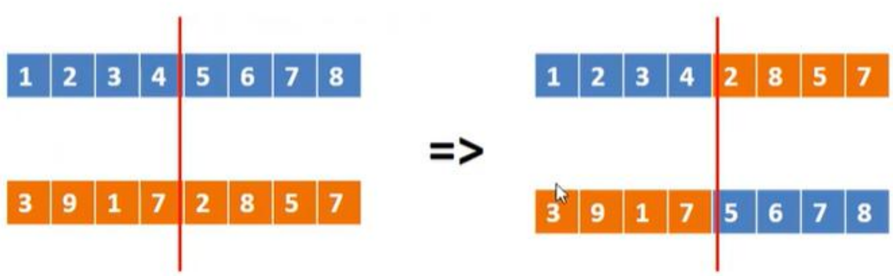


Crossover…

- More than one parent is selected and one or more off-springs are produced using the genetic material of the parents.
- Crossover Operators
  - One Point Crossover
  - Multi Point Crossover
  - Uniform Crossover

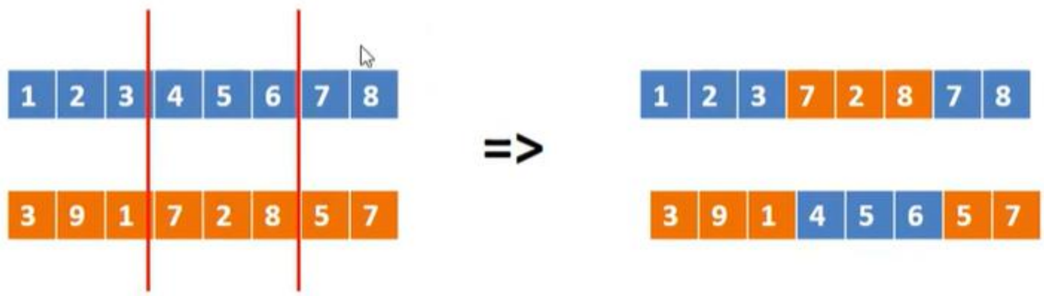
One Point Crossover

- A random crossover point is selected and the tails of its two parents are swapped to get new off-springs.



Multi Point Crossover

- Multi point crossover is a generalization of the one-point crossover wherein alternating segments are swapped to get new off-springs.

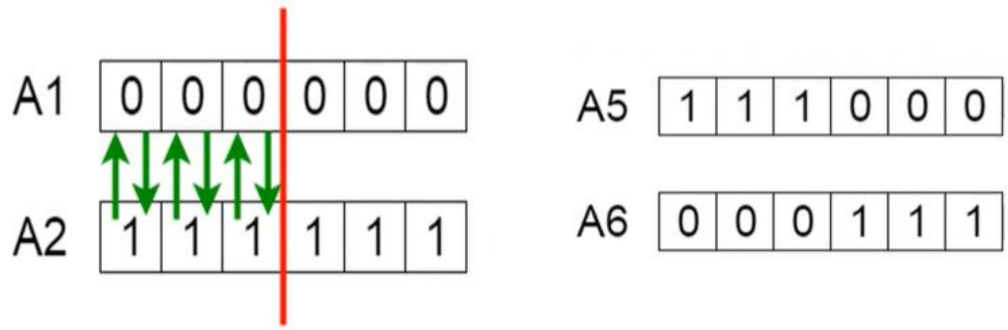


Uniform Crossover

- A coin is flipped for each chromosome to decide whether or not it'll be included in the off-spring.
- Coin can be biased to one parent, to have more genetic material in the child from that parent.

Offspring

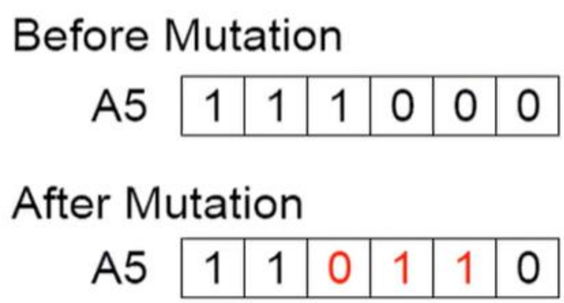
- **Offspring** are created by exchanging the genes of parents among themselves until the crossover point is reached.



- The new offspring are added to the population.

Mutation

- Some of the genes can be subjected to a **mutation** with a low random probability.
- Some of the bits in the bit string can be flipped.



- Mutation occurs to maintain diversity within the population and prevent premature convergence.

Mutation…

- A small random tweak in the chromosome, to get a new solution.
- It is used to maintain and introduce diversity in the genetic population and is usually applied with a low probability.
- Mutation Operators:
  – Bit Flip Mutation
  – Random Resetting
  – Swap Mutation
  – Scramble Mutation
  – Inversion Mutation

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Bit Flip Mutation

- Select one or more random bits and flip them.
- Used for binary encoded GAs.

Random Resetting

- Random Resetting is an extension of the bit flip for the integer representation.
- A random value from the set of permissible values is assigned to a randomly chosen gene.

Swap Mutation

- Select two positions on the chromosome at random, and interchange the values.

3	9	1	7	2	8	5	7	=>	3	9	5	7	2	8	1	7

Scramble Mutation

- From the entire chromosome, a subset of genes is chosen and their values are scrambled or shuffled randomly.

3	9	1	7	2	8	5	7	=>	3	9	8	1	5	7	2	7

Inversion Mutation

- Select a subset of genes like in scramble mutation, but instead of shuffling the subset, merely invert the entire string in the subset.

3	9	1	7	2	8	5	7	=>	3	9	5	8	2	7	1	7

Termination

- The algorithm terminates if the population has converged (does not produce offspring which are significantly different from the previous generation).
- Then it is said that the genetic algorithm has provided a set of solutions to the problem.

Comments

- The population has a fixed size. As new generations are formed, individuals with least fitness die, providing space for new offspring.
- The sequence of phases is repeated to produce individuals in each new generation which are better than the previous generation.

Issues

Choosing basic implementation issues:

- representation
- population size, mutation rate, ...
- selection, deletion policies
- crossover, mutation operators
- Termination Criteria
- Performance, scalability
- Solution is only as good as the evaluation function (often hardest part)

Advantages

- Faster and more efficient as compared to the traditional methods.
- It uses probabilistic transition rules, not deterministic rules.
- Its good for noisy environments.
- Optimizes both continuous and discrete functions and also multi-objective problems.
- Provides a list of “good” solutions and not just a single solution.
- Always gets an answer to the problem, which gets better over the time.
- Its ideal to parallelize the algorithms for implementation
- Useful when the search space is very large and there are a large number of parameters involved.

Basic Structure

1. Randomly generate initial population of n strings (“chromosomes”)
2. Evaluate the fitness of each string in the population
3. Repeat the following steps until next generation of n individual strings produced
   a. Select pair of parent chromosomes from current population according to their fitness, i.e. chromosomes with higher fitness are selected more often
   b. Apply crossover (with probability)
   c. Apply mutation (with probability of occurrence)
4. Apply generational replacement
5. Go to 2 or terminate if termination condition met